

A Robust Scalable Markov Chain Monte Carlo Approach for a Probabilistic Digital Twin of Optical Networks

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Abstract We propose a method of refining the optical network parameters joint probability distribution, leveraging live measurements and integrating a differentiable reimplement of GNPpy into a Bayesian inference framework. We illustrate the benefits of our approach in terms of robustness and quality of the predicted distributions. ©2026 The Author(s)

Introduction

By digital twin (DT) of an optical network we refer to a time-stamped digital replica of the network pertinent physical parameters (PMs) [1]. In most installed networks, quality of transmission (QoT) measurements are available at no extra cost: the pre-FEC bit-error ratio (BER) for each lightpath, or the power at amplifier inputs and outputs. These measurements are rarely sufficient to determine the true value of each PM unambiguously due to degeneracy [2, 3], i.e. due to the fact that different PM combinations can yield identical QoT measurements. Nevertheless, measurements can always be used to update the joint probability distribution function (PDF) of the PMs and reduce uncertainty [4-9] over the PM prior knowledge. A concrete use case of such a DT is brown-field design: knowing which design margins have been consumed allows new lightpaths to be planned with more aggressive design margins. Besides, a practical DT must (i) accept prior PM PDFs of arbitrary form, (ii) provide a full posterior PM PDFs rather than point estimates, and (iii) scale tractably as the number of PMs grows. In [8] we proposed a Markov Chain Monte Carlo (MCMC) approach based on the Metropolis-Hastings (MH) algorithm [10, 11]. This approach satisfies the constraints (i-iii), whose per-sample cost is linear in the number d of PMs. However, MH explores the posterior via an uninformed random walk and becomes increasingly slow to converge as the dimension d increases [8]. State-of-the-art sampling algorithms can address some of these limitations but require the incorporation of gradient information [12-14] from the QoT model, which is usually not available.

In this paper we extend previous work [8], using GNPpy [15] as the basis software for forward QoT calculation from input PMs. We developed a differentiable version of GNPpy's QoT engine, denoted ∂ GNPpy, which enables inference algorithms that exploit the gradient of the log-posterior distribution. In particular, we focus on the No-U-Turn Sampler (NUTS) [14], a self-tuning Hamiltonian Monte Carlo (HMC) variant [13] that scales

substantially better with dimensionality than MH. We also include a Maximum *a posteriori* (MAP) estimator as a computationally light reference, which can be computed via gradient descent [12]. We evaluate MAP, MH and NUTS across topologies of increasing size, different PM subsets, and varying observation types. Our results indicate that NUTS achieves the highest quality of estimated parameter distribution as d grows, without the need of per-experiment tuning.

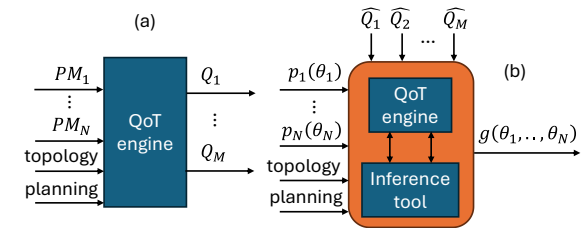


Fig. 1: Block diagram of (a) QoT engine, (b) digital twin

The method

In Fig. 1a we show the conceptual block diagram of the QoT engine, which is ∂ GNPpy in our case. This can be seen as a black box which takes N discrete values of input PMs denoted as PM_k with $k = 1, \dots, N$, the topology of the network and the current planning of established lightpaths. Then the QoT engine calculates the outputs Q_l , $l = 1, \dots, M$ which in our case are the generalized SNRs (GSNRs) for each of the established lightpaths and the total power measurements at the inputs and outputs of amplifiers. In Fig. 1b we show the conceptual diagram of our DT. In this case, the true values of PMs are unknown, but we assume that a collection of prior distributions $p_k(\theta_k)$ with $k = 1, \dots, N$ are available, carrying the prior statistical information of PMs, e.g. restrictions on the PM values coming from physics, datasheets, measurements at the beginning of the system lifetime or the previous (in time) snapshot of the DT database. We further assume that prior distributions are independent, i.e. the joint prior PDF is written as $p_{prior}(\theta) = \prod_{k=1}^N p_k(\theta_k)$ with $\theta = [\theta_1, \dots, \theta_N]$. Live measurements are also fed to the DT, denoted here as $\widehat{Q}_k$, with $k = 1, \dots, M$, and carried by the vector $y =$

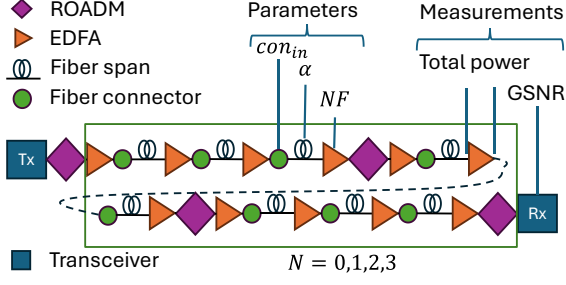


Fig. 2: Point-to-point configuration under test

$[\widehat{Q}_1, \dots, \widehat{Q}_M]$. Then, the DT uses the QoT engine together with the proposed inference tool to reconcile the prior PDFs and the measurements, in order to estimate the joint posterior PDF of the PMs $p_{\text{post}}(\theta|\mathbf{y})$. We note that, $p_{\text{post}}(\theta|\mathbf{y})$ carries all the pertinent information of the digital twin.

To enable gradient-based inference, we re-implemented GNPpy's signal-propagation pipeline as a composition of PyTorch operations [16]. Every element computation — erbium doped fiber amplifier (EDFA) noise figure (NF) application, nonlinearity Gaussian noise (GN) accumulation, reconfigurable optical add-drop multiplexer (ROADM) power equalization — is expressed as a PyTorch module whose parameters (e.g. NF, connector loss, fibre attenuation etc.) can be declared as learnable tensors. Automatic differentiation (autograd) [16] then propagates gradients of any scalar output, e.g. GSNR at the receiver or total power at an intermediate monitor, back through the entire network. The translation from GNPpy to PyTorch was carried out with the assistance of AI coding tools, which substantially accelerated the porting. Finally, to exploit standard probabilistic programming infrastructure, we wrapped ∂ GNPpy into a Pyro [17] model which gives direct access to built-in MCMC inference algorithms (NUTS, in particular). Baseline MH was implemented independently.

Both MH and NUTS produce a Markov chain, which converges to the target stationary joint PDF $g(\theta)$. At each step, MH generates a new candidate sample vector θ' of the PMs by adding isotropic Gaussian noise to the current state θ , i.e. $\theta' = \theta + \sigma_{\text{prop}} \cdot \xi$, $\xi \sim N(0, I)$, where $N(\mu, \sigma^2)$ denotes a multivariate normal distribution, I is the identity matrix and σ_{prop} is a scaling factor. The method accepts or rejects the new sample with probability $\min(1, [p_{\text{post}}(\theta'|\mathbf{y})/p(\theta|\mathbf{y})])$ [10]. Each step requires one gradient-free forward pass through ∂ GNPpy. The scale σ_{prop} is critical and difficult to tune [11]: if too small, exploration is slow; if too large, most proposals are rejected, stalling the chain. Maintaining an appropriate acceptance rate requires careful experiment-specific calibration, and consecutive MH samples often remain strongly

correlated. As a result, the cost of obtaining effectively independent samples typically worsens with dimension d , and under standard assumptions can scale as $O(d)$ [18]. By contrast, NUTS [14], a self-tuning variant of HMC [13], uses the gradient of the log-posterior to propose larger, less correlated moves. By adapting proposals to the local structure of the posterior, HMC-type methods can achieve more favourable scaling than random-walk MH, proportional to $O(d^{1/4})$ in ideal cases, overcompensating the additional cost of gradient evaluations for large d . The enabler of this advantage is ∂ GNPpy: without a differentiable forward model, NUTS cannot be applied.

We also perform preliminary MAP evaluations as a fast proxy to detect potential degeneracies before applying more accurate MCMC methods. MAP estimates the posterior mode by minimizing $L(\theta) = -\log p(\mathbf{y}|\theta) - \log p_{\text{prior}}(\theta)$ via gradient descent, made available by ∂ GNPpy, and provides an uncertainty estimate from the local Hessian around the mode.

Numerical experiments

We evaluate the proposed probabilistic DT on a point-to-point topology shown in Fig. 2. The base chain ($N = 0$) comprises 11 EDFAs, 8 fiber spans, and 4 ROADMs. We replicate the base chain N times to scale the total parameter dimension d . As learnable parameters, we focus on NFs, connector losses con_{in} at fiber inputs and fiber attenuation constants α , while all other parameters are treated as known. For instance, jointly estimating NFs, con_{in} and α at $N = 3$ repetitions yield $d = 11 \times 4 + 8 \times 4 + 8 \times 4 = 108$ parameters.

Three WDM channels are co-propagated at $f \in \{193.04, 193.41, 193.79\}$ THz with a symbol rate of 33 GBd. GSNRs are collected for a launched power $P = 3$ dBm, which corresponds to the optimal GSNR for the given configuration. GSNR and power measurements are assumed to be normally distributed with standard deviation equal to 0.1 dB and 0.3 dB respectively.

Total-power readings at the output of the 3 intermediate ROADMs or additional per-EDFA total in/out power readings supplement the GSNR measurement and help to reduce parameter degeneracy. Synthetic ground-truth parameters are drawn as follows: NFs centred at 4.0 ± 0.15 dB, connector losses at 0.5 ± 0.5 dB, and attenuation constants at 0.220 ± 0.03 dB/km. All priors over PMs are independent truncated Gaussians $TN(\mu, \sigma; [l, h])$: NFs have $\mu \in [4.9, 5.9]$ and $\sigma = 1.41$ dB truncated in $[3, 7]$ dB; con_{in} has prior $p_{\text{con}_{\text{in}}, \text{prior}} \sim TN(0.0, 0.45; [0, 2])$ dB; α has prior $p_{\alpha, \text{prior}} \sim TN(0.22, 0.045; [0.14, 0.30])$ dB/km.

We run a grid of posterior-estimation

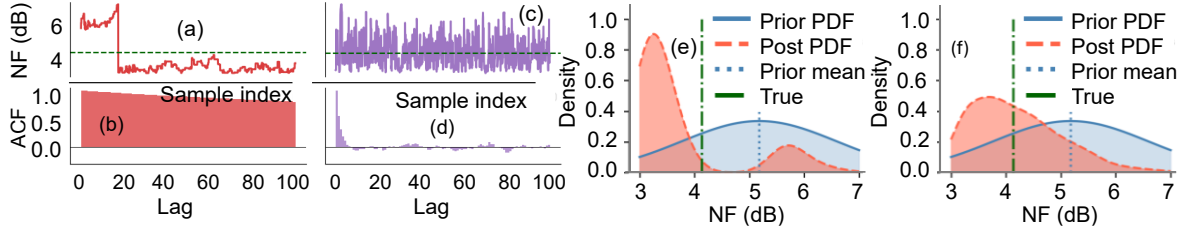


Fig. 3: MCMC trace and ACF (lag 200) for (a-b) MH and (c-d) NUTS, illustrating the EDFA1 NF in the ($N=3$) setting with joint inference of noise figures and connector losses. Relative prior and posterior PDFs are shown for (e) MH and (f) NUTS.

experiments with MAP, MH, and NUTS over different measurement types (GSNR and power) and parameter subsets (NFs , con_{in} and α). MH and NUTS use 1000 warm-up (burn-in) iterations and 1500 post-warm-up samples. MH employs a per-parameter Gaussian proposal, calibrated during warm-up to maintain an appropriate acceptance rate [14]. NUTS adapts step size and mass matrix during warm-up. Each experiment is repeated five times with different random seeds.

We first examine the Markov chain quality for $N = 3$, jointly inferring NFs and con_{in} from GSNRs and power measurements. We show the sample traces and the autocorrelation as a function of lag for MH (Fig. 3a, b) and NUTS (Fig. 3c, d). In this setting, MH suffers from strong sample-to-sample correlation and frequent proposal rejections, while NUTS mixes rapidly and produces nearly uncorrelated samples. The resulting posterior PDFs, shown for MH in Fig. 3e and for NUTS in Fig. 3f, confirm this: the apparent multimodality under MH is an artifact of poor mixing, whereas the NUTS estimate is broader and better centred on the ground truth.

Fig. 4a illustrates the effect of observation type on the MAP estimate of con_{in} , with shaded error-bars depicting the uncertainty assessment of MAP: GSNR alone leaves the degeneracy unresolved and the estimate collapses to the prior mean, while adding power measurements concentrates it around the ground truth. Fig. 4b shows the mean MAE across runs on connector losses versus the number of inferred parameters d , with shaded error-bars depicting the standard deviation. The results correspond to a joint estimation of NFs , con_{in} and α using GSNR and power measurements. NUTS consistently outperforms MH with lower MAE, which in turn outperforms MAP, while NUTS also shows smaller standard deviation, i.e. greater consistency

across runs. In the same setting, we show the absolute correlation matrix of all parameters for $N = 0$, with each parameter type indicated by the corresponding tag for MH (in Fig. 4c) and NUTS (Fig. 4d). We note that, generally, correlation between PMs is high for MH, and low for NUTS, with the exception of connector losses and attenuation constants, for which degeneracy is intrinsically high. This is yet another proof of the robustness of NUTS to efficiently disentangle the probability distributions of different PMs. Brown-field planning can then be performed with (optionally) correlated PMs, as e.g. discussed in [9].

These observations generalise across all joint-estimation experiments: NUTS consistently delivers well-mixed posteriors without manual tuning, especially when both GSNR and power measurements are available. When only GSNRs are observed, the data does not carry enough information to resolve parameter degeneracy, and the posterior correctly collapses toward the prior, a behaviour that NUTS captures. MH is more sensitive to the experimental setting: it performs adequately in simpler configurations but becomes unreliable as network complexity grows.

Conclusion

We propose and evaluate the use of NUTS algorithm, leveraging a differentiable version of GNPpy, for the development of a scalable probabilistic digital twin of optical networks. We show that the approach improves upon previously proposed MH-based method by generating higher-quality samples with lower autocorrelation and by reducing the risk of the sampler becoming trapped in poorly explored regions of the parameter space. At the same time, it preserves favourable computational scaling as the number of PM increases. We note that the method is compatible with any differentiable QoT engine.

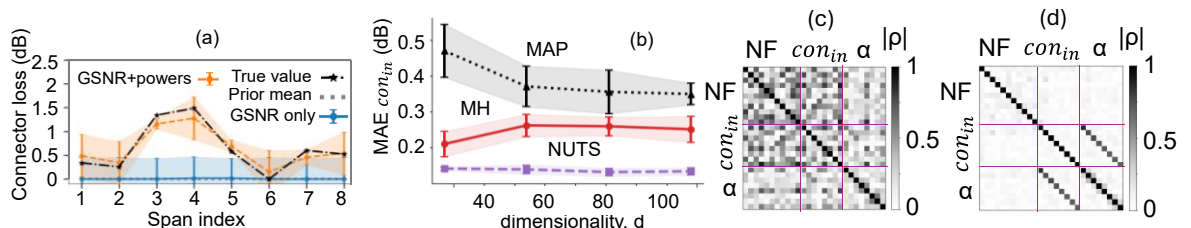


Fig. 4: (a) MAP estimates of con_{in} , for $N = 0$. (b) MAE vs. dimensionality d for the 3 explored algorithms for joint inference of all parameters; corresponding absolute value of Pearson correlation for (c) MH and (d) NUTS ($N = 0$).

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